indicates a maximum propagation delay of less than 50ns.

Improved common-mode transient immunity

Isolated gate drivers are very widely used in solar inverters, electric vehicles (EVs), network power, etc. The isolated gate drivers, in addition to driving MOSFETs or IGBTs, provide galvanic isolation. Common-mode transient immunity (CMTI) is an important parameter for high-frequency isolated gate drivers as they deal with a differential voltage between two separate ground references.

CMTI is defined as the maximum tolerable rate of rising or fall of the common-mode voltage applied. In other words, CMTI dictates how fast common-mode transients can be subjected. It is measured in kV/µs or V/ns, and high CMTI means that the two isolated circuits function correctly as per the specifications.

Infineon EiceDRIVER isolated gate drivers feature the highest CMTI of over 200kV/µs. TI's isolated gate drivers UCC21710 and UCC21759-Q1 provide >150V/ns CMTI. Another isolated gate driver by TI, UCC23514, features a minimum 150kV/µs CMTI.

Soft turn-off and Miller clamp

The desaturation (DESAT) fault detection feature in gate drivers protects power switches against short-circuit events. During such fault events, turning the power switch off too quickly can generate a high voltage spike at the power switch collector. Therefore, most of the latest gate drivers employ a soft turn-off sequence. The soft turn-off is done by increasing the gate voltage discharge time during the turn-off to reduce the rapid current changes after DESAT faults.

Miller clamp is not a new technique in gate drivers, but it is commonly used in the latest SiC gate drivers. One of the common problems faced when driving power switches like IGBTs is parasitic turn-on due to the Miller capacitor. Therefore, a Miller clamp is used that supplies the gate with a lower impedance path to reduce the chance of Miller capacitance induced turn-on.

Smart gate drive architecture

Smart gate drive is an exclusive feature offered by Texas Instruments for enhancing gate drive operation to drive an electric motor. The feature allows more control over external MOSFETs, providing design engineers flexibility. This new architecture manages dead time to prevent shoot-through, controls slew rate to decrease electromagnetic interference (EMI), and optimises propagation delay for optimal performance.

TI's DRV8706-Q1, DRV8705-Q1, DRV871x-Q1, and DRV8106 gate drivers feature this smart gate drive architecture.

Conclusion

Gate drivers play an important role in rapid electrification of automobiles and development of renewable energy systems. As the demand for high-efficiency power converters is increasing, there arises the need for high-performance gate driver ICs for various power electronics applications. EFY

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